

STATE BOARD OF TECHNICAL EDUCATION AND TRAINING
TELANGANA

DIPLOMA EXAMINATION (C-24)

C-24-OCT-2025

SEMESTER III, MID-II EXAM

AA/AI/AU/BM/CCB/CE/CH/Common/CPS/CS/EC/EE/EI/ES/EV/LF/LG/ME/MN/MT/PK/PT/TT

SC-301

Applied Engineering Mathematics (Open Book System)



PCODE
243001

Exam Date: 16-10-2025

Session: FN

Duration: 1 Hour [09:30 AM To 10:30 AM]

[Total Marks: 20]

PART-A

Instructions:

1. Answer the following questions.

4 X 1 = 4

2. Each question carries **ONE** mark.

1. Write the first and second functions while applying integration by parts to

evaluate $\int x^2 \sin\left(\frac{\pi x}{7}\right) dx.$

2. Evaluate $\int e^x \sin 2x dx.$

3. Evaluate the integral $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin x dx.$

4. Find the value of the integral $\int_0^1 e^{-x} dx.$

PART-B

Instructions:

1. Answer the following questions.

2 X 3 = 6

2. Each question carries **THREE** marks.

- 5(a). Evaluate $\int e^x [\tan x + \log(\sec x)] dx.$

----- OR -----

- 5(b). Evaluate $\int x e^{-5x} dx.$

- 6(a). Evaluate $\int_1^e \log x dx.$

----- OR -----

- 6(b).

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Evaluate $\int_0^{\pi} \sin^3 x \, dx.$

PART-C

Instructions:

1. Answer the following questions.
2. Each question carries **FIVE** marks.

$2 \times 5 = 10$

7(a). Evaluate $\int x^3 \cos 5x \, dx$ by using Bernoulli's rule.

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---- OR ----

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7(b). Evaluate $\int \frac{2}{(1-x)(1+x^2)} \, dx.$

8(a). Evaluate $\int_0^{\frac{\pi}{2}} \frac{\sin x}{\sin x + \cos x} \, dx.$

---- OR ----

8(b). Evaluate $\int_0^{\pi/4} x \cdot \sec^2 x \, dx.$

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